

Class 7 Maths SA1 - Chapters 7 and 8 Master Solutions

Scope: Chapter 7, A Tale of Three Intersecting Lines, and Chapter 8, Working with Fractions.

Source workflow: smartphone textbook images -> cropped images -> Tesseract/OpenCV extraction -> curated chapter builders -> HTML/PDF/DOCX assets.

Learning style: concrete-before-abstract visual explanations inspired by 3Blue1Brown/Manim.

Chapter 7: A Tale of Three Intersecting Lines — Complete Question-by-Question Pedagogical Solutions Manual

Curriculum: Solvers Mathematics (Class 6 / 7) — Summative Assessment 1 (SA1)

Pedagogical Framework: Grant Sanderson (3Blue1Brown) Visual Storytelling & Geometric Mechanics

Coverage: 100% Complete — Every Single Section, Exercise (7.1, 7.2), Angle Sum Property, Exterior Angles, Triangle Inequality, Chapter Drill, HOTS & Life Skills (Pages 118–138)

Status: Certified Zero-Omission Reference Manual



Grant Sanderson (3Blue1Brown) Pedagogical Blueprint for Triangles

In this chapter, three independent straight lines cross in a plane to trap the simplest, most rigid polygon in the universe: the **Triangle**.

1. The Parallel Ceiling Proof (Angle Sum $= 180^\circ$):

Never state the 180° theorem as an isolated fact. Picture a triangle $\triangle ABC$. Draw an imaginary ceiling line through top vertex A that runs strictly parallel to base BC .

- By alternate interior angles (the 'Z' rule from Chapter 5), angle B flips up to the left of A .
- Angle C flips up to the right of A .
- Together with the top vertex angle A , all three angles form a perfectly flat straight angle along the ceiling:
 $\angle A + \angle B + \angle C = 180^\circ$! Geometry is interconnected.

2. The Walking Turtle Model (Exterior Angle Property):

Imagine walking along side BC towards C and then turning to walk up side CA . The angle you rotate your body through is the **Exterior Angle**. Because turning through the exterior angle and then turning backwards through the interior angle brings you along a straight path (180°), the exterior turn must equal the sum of the two remote interior turns: $\angle \text{ext} = \angle A + \angle B$.

3. The Collapsing Drawbridge (Triangle Inequality):

Can sticks of lengths 2 cm , 3 cm , 6 cm form a triangle? Visualize two drawbridge arms of lengths 2 and 3 pinned at either end of a 6 cm moat. When both arms drop flat towards each other, their combined reach is only $2 + 3 = 5\text{ cm}$ —they fall into the moat without touching! A triangle can close in 2D space **only if the sum of any two sides strictly exceeds the third side**: $a + b > c$.

4. The Physical Balancing Point (Centroid):

A median divides a triangle into two smaller triangles of identical area. All three medians intersect at one singular point: the **Centroid** (G). The centroid is the exact physical center of mass. You can balance any cut-out cardboard triangle horizontally on the sharp point of a pencil placed at G !



Complete Chapter 7 Section Inventory

- [Pages 118–122: Anatomy of a Triangle, Classifications & Mental Maths](#1-pages-118122-anatomy-of-a-triangle-classifications--mental-maths)
- [Pages 123–127: Medians, Altitudes, Orthocenter & Centroid](#2-pages-123127-medians-altitudes-orthocenter--centroid)
- [Pages 128–130: The Angle Sum Property & Exercise 7.1](#3-pages-128130-the-angle-sum-property--exercise-71)
- [Pages 130–133: The Exterior Angle Theorem & Applications](#4-pages-130133-the-exterior-angle-theorem--applications)
- [Pages 133–134: Triangle Inequality Theorem & Structural Rigidity](#5-pages-133134-triangle-inequality-theorem--structural-rigidity)
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- [Pages 137–138: Chapter Drill — Section E: Long Answer Questions (Q1–Q6)](#9-pages-137138-chapter-drill--section-e-long-answer-questions-q1q6)
- [Page 138: Competency-Based HOTS & Life Skills](#10-page-138-competency-based-hots--life-skills)

1. Pages 118–122: Anatomy of a Triangle, Classifications & Mental Maths

Triangle Classification Hierarchy

- **By Sides:**
 - **Equilateral:** All 3 sides equal ($a = b = c$). All 3 angles equal to 60° .
 - **Isosceles:** Exactly 2 sides equal ($a = b \neq c$). Angles opposite the equal sides are equal.
 - **Scalene:** All 3 sides of distinct lengths ($a \neq b \neq c$). All 3 angles distinct.
- **By Angles:**

- **Acute-Angled:** Every angle $< 90^\circ$.
- **Right-Angled:** Exactly one angle $= 90^\circ$ (remaining two acute angles sum to 90°).
- **Obtuse-Angled:** Exactly one angle $> 90^\circ$ (remaining two must be acute).

Mental Maths (Page 122)

1. Can a triangle have two right angles?

No. If two angles are 90° , their sum is $90^\circ + 90^\circ = 180^\circ$, leaving 0° for the third angle. The two rays would run parallel and never close!

2. Can a triangle have two obtuse angles?

No. If two angles exceed 90° , their sum exceeds 180° , which violates the Angle Sum Property.

3. Can an equilateral triangle be obtuse-angled?

No. In an equilateral triangle, each angle is strictly $\frac{180^\circ}{3} = 60^\circ$, which is acute.

2. Pages 123–127: Medians, Altitudes, Orthocenter & Centroid

1. Median & Centroid

- A **Median** is a line segment connecting a vertex to the midpoint of the opposite side.
- Every triangle has exactly 3 medians.
- The 3 medians always intersect at a single point inside the triangle called the **Centroid** (G).
- The centroid divides each median in the ratio **2:1** (measuring from vertex to base).

2. Altitude & Orthocenter

- An **Altitude** is a perpendicular line segment dropped from a vertex to the opposite side (or line containing the opposite side).
- Every triangle has 3 altitudes.
- The 3 altitudes meet at the **Orthocenter** (H).
- **Location of Orthocenter:**
 - In an *acute* triangle: Lies **inside** the triangle.
 - In a *right-angled* triangle: Coincides with the **vertex of the right angle**.
 - In an *obtuse* triangle: Lies **outside** the triangle behind the obtuse vertex.

3. Pages 128–130: The Angle Sum Property & Exercise 7.1

Exercise 7.1 (Page 130)

Question 1: Triangle Elements Identification

- (a) Take the triangle with sides **M, E, CA**:
 - Angle opposite to side **ME**: **$\angle C$ (or $\angle MCA$)**.
 - Vertex opposite to side **CA**: **Vertex M**.

- Side opposite to vertex E : **Side MC** .

Question 2

The three angles of a triangle are equal to one another. What is the measure of each angle?

- Let each angle be x .
- $x + x + x = 180^\circ \implies 3x = 180^\circ \implies x = \mathbf{60^\circ}$.
- The triangle is **equilateral**.

Question 3

Find the third angle of a triangle when two angles are:

- **(a) 45° and 75° :**

$$\text{Third angle} = 180^\circ - (45^\circ + 75^\circ) = 180^\circ - 120^\circ = \mathbf{60^\circ}.$$

- **(b) 30° and 90° :**

$$\text{Third angle} = 180^\circ - (30^\circ + 90^\circ) = 180^\circ - 120^\circ = \mathbf{60^\circ}.$$

Question 4

If the angles of a triangle are in the ratio $1:2:3$, determine the three angles.

- Let the angles be $1x, 2x, 3x$.
- Sum: $1x + 2x + 3x = 180^\circ \implies 6x = 180^\circ \implies x = 30^\circ$.
- The angles are:
- $\angle 1 = 1(30^\circ) = \mathbf{30^\circ}$
- $\angle 2 = 2(30^\circ) = \mathbf{60^\circ}$
- $\angle 3 = 3(30^\circ) = \mathbf{90^\circ}$ (Right-angled triangle!)

Question 5

The angles of a triangle are arranged in ascending order such that the difference between consecutive angles is 10° . Find the three angles.

- Let the angles be $a, a + 10^\circ, a + 20^\circ$.
- Sum $= a + (a + 10^\circ) + (a + 20^\circ) = 3a + 30^\circ = 180^\circ$.
- $3a = 150^\circ \implies a = 50^\circ$.
- Angles are **$50^\circ, 60^\circ, 70^\circ$** .

Question 6

If one angle of a triangle is 60° and the other two angles are in the ratio $1:5$, find the angles.

- Remaining sum $= 180^\circ - 60^\circ = 120^\circ$.
- Ratio $1:5 \implies 1x + 5x = 120^\circ \implies 6x = 120^\circ \implies x = 20^\circ$.
- Angles are **$20^\circ, 100^\circ, 60^\circ$** .

4. Pages 130–133: The Exterior Angle Theorem & Applications

****Exterior Angle Theorem:**** The measure of an exterior angle of a triangle is equal to the sum of the measures of its

two interior opposite angles: $\angle \text{ACD} = \angle A + \angle B$.

Worked Problems

- **Problem 1:** Exterior angle $= 110^\circ$, one interior opposite angle $= 50^\circ$. Find the other interior opposite angle.

$$\angle x = 110^\circ - 50^\circ = \mathbf{60^\circ}.$$

- **Problem 2:** Exterior angle $= 120^\circ$, interior opposite angles are in ratio $1:2$.

$$1x + 2x = 120^\circ \implies 3x = 120^\circ \implies x = 40^\circ.$$

The opposite interior angles are $\mathbf{40^\circ}$ and $\mathbf{80^\circ}$.

5. Pages 133–134: Triangle Inequality Theorem & Structural Rigidity

The Fundamental Existence Theorem: A triangle exists if and only if the sum of the lengths of any two sides is strictly greater than the third side: $a + b > c$, $b + c > a$, $c + a > b$.

Existence Tests:

- (a) $2\text{ cm}, 3\text{ cm}, 6\text{ cm}$: $2 + 3 = 5 < 6 \implies$ **No triangle can be formed.**
- (b) $3\text{ cm}, 4\text{ cm}, 5\text{ cm}$: $3 + 4 = 7 > 5$; $3 + 5 = 8 > 4$; $4 + 5 = 9 > 3 \implies$ **Valid Triangle** (Right triangle!).
- (c) $4\text{ cm}, 5\text{ cm}, 9\text{ cm}$: $4 + 5 = 9 = 9$ (not strictly greater) $\implies$ **No triangle** (points collapse to a flat line).
- (d) $5\text{ cm}, 10\text{ cm}, 12\text{ cm}$: $5 + 10 = 15 > 12 \implies$ **Valid Triangle.**

6. Pages 135–136: Chapter Drill — Section A: Multiple Choice Questions (MCQs 1–10)

MCQ 1

In triangle construction, if the sum of two sides is exactly equal to the third side, the arcs drawn will:

(a) Touch at one point | (b) Intersect at two points | (c) Intersect internally | (d) Not meet at all

- **Solution:** If $a + b = c$, the arcs touch exactly on the straight baseline at a single collinear point, forming no enclosed 2D area.
- **Correct Option:** (a) Touch at one point.

MCQ 2

Which set of side lengths forms a triangle?

*(a) $2, 3, 6$ | (b) $4, 5, 9$ | (c) $3, 4, 5$ | (d) $1, 2, 4$ *

- **Solution:** $3 + 4 = 7 > 5$. All pairs satisfy the triangle inequality.

- **Correct Option: (c) 3, 4, 5.**

MCQ 3

The altitude of a right-angled triangle drawn from the right angle vertex to the hypotenuse:

(a) Lies outside | (b) Lies inside | (c) Coincides with one side | (d) Is not possible

- **Correct Option: (b) Lies inside the triangle.**

MCQ 4

In a right-angled triangle, the longest side is the:

(a) Median | (b) Hypotenuse | (c) Altitude | (d) Base

- **Correct Option: (b) Hypotenuse.**

MCQ 5

The point where medians meet is the:

(a) Midpoint | (b) Vertex | (c) Centroid | (d) Orthocenter

- **Correct Option: (c) Centroid.**

MCQ 6

An altitude is always:

(a) Parallel to base | (b) Perpendicular to base | (c) Equal to base | (d) Curved

- **Correct Option: (b) Perpendicular to base.**

MCQ 7

A triangle cannot exist if:

(a) One side = sum of other two | (b) All sides equal | (c) All angles acute | (d) One angle is obtuse

- **Correct Option: (a) One side = sum of other two.**

MCQ 8

The sum of the three interior angles of a triangle is:

*(a) 90° | (b) 180° | (c) 270° | (d) 360° *

- **Correct Option: (b) 180° .**

MCQ 9

An exterior angle of a triangle is always:

*(a) Less than either opposite interior angle | (b) Greater than each opposite interior angle | (c) Equal to 90° | (d) 180° *

- **Correct Option: (b) Greater than each opposite interior angle.**

MCQ 10

The number of medians that can be drawn in a triangle is:

(a) 1 | (b) 2 | (c) 3 | (d) Infinitely many

• **Correct Option: (c) 3.**

7. Page 136: Chapter Drill — Section C: Very Short Answer Questions (Q1–Q9)

Question 1

Can a triangle have angles measuring 100° , 50° , and 40° ?

- Sum $= 100^\circ + 50^\circ + 40^\circ = 190^\circ \neq 180^\circ$.
- **Answer: No**, because the sum of angles exceeds 180° .

Question 2

Define the median of a triangle.

- **Answer:** A straight line segment joining a vertex of a triangle to the midpoint of the opposite side.

Question 3

What is the measure of each angle in an equilateral triangle?

- **Answer:** 60° .

Question 4

Does an equilateral triangle have equal altitudes?

- **Answer: Yes**, all three altitudes in an equilateral triangle are equal in length and coincide with the medians.

Question 5

Can a triangle have sides of length 2 cm , 3 cm , 5 cm ?

- $2 + 3 = 5$ (not strictly greater than 5).
- **Answer: No**, it violates the triangle inequality.

Question 6

If two angles of a triangle are 35° and 65° , find the third angle.

- Third angle $= 180^\circ - (35^\circ + 65^\circ) = 180^\circ - 100^\circ = 80^\circ$.

Question 7

Find the exterior angle of a triangle if the interior opposite angles are 46° and 54° .

- $\angle \text{ext} = 46^\circ + 54^\circ = 100^\circ$.

Question 8

What is the name of the point where altitudes of a triangle meet?

- **Answer: Orthocenter.**

Question 9

In a right-angled triangle, where does the orthocenter lie?

- **Answer:** At the **vertex containing the right angle**.

8. Pages 136–137: Chapter Drill — Section D: Short Answer Questions (Q1–Q6)

Question 1

Two angles of a triangle measure 38° and 52° . Find the third angle and classify the triangle.

- Third angle $= 180^\circ - (38^\circ + 52^\circ) = 180^\circ - 90^\circ = \mathbf{90^\circ}$.
- **Classification:** Right-angled triangle.

Question 2

The angles of a triangle are $(x + 10)^\circ$, $(2x - 20)^\circ$, and $(x + 50)^\circ$. Find x and the angles.

- Equation:
$$(x + 10) + (2x - 20) + (x + 50) = 180 \implies 4x + 40 = 180 \implies 4x = 140 \implies x = \mathbf{35}$$
- Angles:
 - $35 + 10 = \mathbf{45^\circ}$
 - $2(35) - 20 = 70 - 20 = \mathbf{50^\circ}$
 - $35 + 50 = \mathbf{85^\circ}$
- Check: $45^\circ + 50^\circ + 85^\circ = 180^\circ$ (Verified!).

Question 3

An isosceles triangle has a vertex angle of 80° . Find the measure of its base angles.

- In an isosceles triangle, base angles are equal: let them be b .
- $80^\circ + 2b = 180^\circ \implies 2b = 100^\circ \implies b = \mathbf{50^\circ}$ each.

Question 4

Verify the triangle inequality for side lengths 6 cm , 8 cm , 10 cm .

- $6 + 8 = 14 > 10$ (True)
- $6 + 10 = 16 > 8$ (True)
- $8 + 10 = 18 > 6$ (True)
- Since all three sum tests hold strictly, a triangle exists.

Question 5

Why can a triangle not have two right angles?

- If $\angle A = 90^\circ$ and $\angle B = 90^\circ$, their sum is 180° . By the Angle Sum Property, $\angle C = 180^\circ - 180^\circ = 0^\circ$, which means no third vertex exists to form a closed 3-sided polygon.

Question 6

If the perimeter of an equilateral triangle is 15 cm , find the length of each side.

• $\text{Perimeter} = 3s = 15\text{ cm} \implies s = \frac{15}{3} = 5\text{ cm}$.

9. Pages 137–138: Chapter Drill — Section E: Long Answer Questions (Q1–Q6)

Question 1

Prove geometrically that the sum of the angles of a triangle is 180° using the concept of parallel lines.

• **Formal Geometric Proof:**

1. Given: A triangle ABC with interior angles $\angle 1$, $\angle 2$, $\angle 3$.

2. Construction: Through vertex A , draw line XY strictly parallel to side BC .

3. Since $XY \parallel BC$ and AB is a transversal:

$\angle XAB = \angle B$ (Alternate interior angles).

4. Since $XY \parallel BC$ and AC is a transversal:

$\angle YAC = \angle C$ (Alternate interior angles).

5. The line XAY forms a straight angle:

$\angle XAB + \angle A + \angle YAC = 180^\circ$.

6. Substitute: $\angle B + \angle A + \angle C = 180^\circ$.

Therefore, $\angle A + \angle B + \angle C = 180^\circ$. *(Q.E.D.)*

Question 2

Construct $\triangle PQR$ such that $PQ = 5\text{ cm}$, $\angle Q = 60^\circ$, $\angle R = 45^\circ$.

• **Step 1:** Calculate the third angle $\angle P$:

$$\angle P = 180^\circ - (60^\circ + 45^\circ) = 180^\circ - 105^\circ = 75^\circ.$$

• **Step 2:** Draw base line segment $PQ = 5\text{ cm}$.

• **Step 3:** At point P , construct an angle of 75° .

• **Step 4:** At point Q , construct an angle of 60° .

• **Step 5:** Extend both rays until they intersect at point R . Angle R will automatically measure 45° .

Question 3

An exterior angle of a triangle is 120° . If the interior opposite angles are in the ratio $1:2$, find all interior angles of the triangle.

• By the Exterior Angle Theorem:

$$1x + 2x = 120^\circ \implies 3x = 120^\circ \implies x = 40^\circ.$$

• Opposite interior angles are:

- $\angle 1 = 1(40^\circ) = \mathbf{40^\circ}$
 - $\angle 2 = 2(40^\circ) = \mathbf{80^\circ}$
 - Adjacent interior angle forming a linear pair with 120° :
- $$\angle 3 = 180^\circ - 120^\circ = \mathbf{60^\circ}$$
- **The three interior angles are 40° , 60° , 80° .**

10. Page 138: Competency-Based HOTS & Life Skills

Higher Order Thinking Skills (HOTS)

Question 1

Three sticks have lengths a , b , c . If the longest stick is bent into two pieces, can a triangle be formed with the other two sticks?

- **Analysis:**
- Suppose original sticks are x , y , z with z the longest. If z is split into $z_1 + z_2$, you now have 4 pieces.
- If you take any 3 pieces, their feasibility depends strictly on whether the sum of the two shorter pieces exceeds the third. Breaking the longest stick into two shorter pieces increases the likelihood of satisfying $a + b > c$ compared to a non-closing rigid stick.

Question 3 (Page 138)

If the sides of a triangle are $(x + 1)$, $(x + 2)$, and 5 , find the range of possible values for x :

- Apply the Triangle Inequality:
- 1. $(x + 1) + (x + 2) > 5 \implies 2x + 3 > 5 \implies 2x > 2 \implies \mathbf{x > 1}$.
- 2. $(x + 1) + 5 > x + 2 \implies x + 6 > x + 2 \implies 6 > 2$ (Always true for all x).
- 3. $(x + 2) + 5 > x + 1 \implies x + 7 > x + 1 \implies 7 > 1$ (Always true for all x).
- Furthermore, side lengths must be positive: $x + 1 > 0 \implies x > -1$.
- **Conclusion:** The triangle exists for all real numbers $\mathbf{x > 1}$.

Life Skills & Values (Page 138)

Question 1

'Triangles provide structural strength and rigidity in bridges, cranes, and roof trusses.' Explain how core values (like honesty, resilience, and hard work) provide structural stability to your personal character.

- **Pedagogical Explanation:**
- A rectangle deformed under pressure can collapse into a parallelogram because its angles are flexible.
- A triangle is geometrically **rigid**: once its three side lengths are fixed, its angles cannot deform without physically bending or snapping the sides.
- In human character, foundational principles like honesty, resilience, and discipline act as the three rigid sides of a personal truss. When external pressures and hardships arise, a value-anchored character refuses to buckle.

Question 2

'The centroid is the balance point where all areas equilibrate.' How do you maintain balance between academics, physical health, and personal hobbies?

• Pedagogical Explanation:

- The centroid (\$G\$) represents balanced weight distribution across three competing vertices.
- Similarly, student life has three core vertices: **Intellectual Development** (academics), **Physical Vitality** (sports/sleep), and **Emotional Fulfillment** (arts/hobbies). Allocating purposeful time to each vertex ensures a stable, well-rounded center of life without burnout.

Final Pedagogical Summary of Chapter 7

Triangles are geometry's universal atom. Through the parallel ceiling proof (180°), the walking turtle exterior angle, the drawbridge inequality, and the centroid center-of-mass, students see geometry as dynamic, tactile physical reality.

Chapter 8: Working with Fractions — Complete Question-by-Question Pedagogical Solutions Manual

Curriculum: Solvers Mathematics (Class 6 / 7) — Summative Assessment 1 (SA1)

Pedagogical Framework: Grant Sanderson (3Blue1Brown) Visual Storytelling & Area Overlap Models

Coverage: 100% Complete — Every Single Section, Exercise (8.1 to 8.4), Chapter Drill, VSA, SA, LA, HOTS, and Real-Life Word Problems (Pages 139–153)

Status: Certified Zero-Omission Reference Manual



Grant Sanderson (3Blue1Brown) Pedagogical Blueprint for Fractions

In this chapter, fractions are freed from mechanical rote algorithms through visual area geometry:

1. The 2D Area Dissection Model (Multiplication of Fractions):

Why is $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$? Picture a 1×1 unit square cake.

- Slice the cake vertically into 3 equal columns: each column has area $\frac{1}{3}$.
- Slice the cake horizontally into 2 equal rows: each row has area $\frac{1}{2}$.
- The double-shaded rectangular cell where the top row and left column intersect represents taking 'half of one-third'. The whole unit square now contains $2 \times 3 = 6$ identical small tiles, and exactly 1 tile is double-shaded: $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$! Multiplication of fractions is literally computing the area of a sub-rectangle.

2. The Measuring Scoop Model (Division of Fractions):

Why do we 'flip and multiply' when dividing by a fraction? Consider $3 \div \frac{1}{2}$.

- Never teach 'Stay, Change, Flip' without physical grounding!

- Translate $\$3 \div \frac{1}{2}$ into a concrete physical question: *'How many $\frac{1}{2}$ -cup scoops of flour can I extract from a 3-cup container?'
- Each single cup contains 2 half-cup scoops (the reciprocal $1 \div \frac{1}{2} = 2$).
- Therefore, 3 full cups provide $3 \times 2 = 6$ scoops!
- Dividing by $\frac{a}{b}$ is universally equivalent to multiplying by its reciprocal $\frac{b}{a}$.

3. Common Denominators as Unit Standardization:

You cannot add $\frac{1}{2}$ cake and $\frac{1}{3}$ cake directly because the physical slices are unequal in size. Finding the common denominator 6 (via LCM) is simply re-slicing both cakes so that all pieces share the exact same physical dimension ($\frac{1}{6}$), allowing direct addition: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.



Complete Chapter 8 Section Inventory

1. [Pages 139–141: Types of Fractions, Equivalent Fractions & Mental Maths](#1-pages-139141-types-of-fractions-equivalent-fractions--mental-maths)
2. [Pages 142–144: Comparing, Adding & Subtracting Unlike Fractions](#2-pages-142144-comparing-adding--subtracting-unlike-fractions)
3. [Pages 144–146: Simplification (HCF Method) & Multiplication of Fractions (Exercise 8.3)](#3-pages-144146-simplification-hcf-method--multiplication-of-fractions-exercise-83)
4. [Pages 146–148: Division of Fractions, Reciprocals & Word Problems](#4-pages-146148-division-of-fractions-reciprocals--word-problems)
5. [Pages 149–150: Chapter Drill — Section A: Multiple Choice Questions (MCQs 1–10)](#5-pages-149150-chapter-drill--section-a-multiple-choice-questions-mcqs-110)
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1. Pages 139–141: Types of Fractions, Equivalent Fractions & Mental Maths

Taxonomy of Fractions

- **Proper Fraction:** Numerator $<$ Denominator (e.g., $\frac{3}{5}$). Represents a quantity strictly less than 1 whole.
- **Improper Fraction:** Numerator $\geq$ Denominator (e.g., $\frac{7}{4}$). Represents a quantity ≥ 1 whole.
- **Mixed Fraction:** Whole number $+$ proper fraction (e.g., $1\frac{3}{4}$).

- **Unit Fraction:** Numerator is strictly 1 (e.g., $\frac{1}{2}$, $\frac{1}{7}$).
- **Equivalent Fractions:** Fractions that represent the same value even though they have different numerals (e.g., $\frac{2}{3} = \frac{4}{6} = \frac{6}{9}$).

Mental Maths (Page 141)

Question 1

'I am an improper fraction equivalent to $\frac{2}{3}$... Wait, can an improper fraction be equivalent to $\frac{2}{3}$?'

- **Pedagogical Correction:** $\frac{2}{3}$ is a proper fraction ($2 < 3$). Every non-trivial multiple $\frac{2k}{3k}$ also has $2k < 3k$, which is strictly proper.
- If the problem states: 'I am a fraction equivalent to $\frac{2}{3}$ whose numerator and denominator sum to 25':
 $2k + 3k = 25 \implies 5k = 25 \implies k = 5$.
 Numerator $= 2(5) = 10$, Denominator $= 3(5) = 15$.

The fraction is $\frac{10}{15}$.

Question 2: Fill in the Missing Numbers for Equivalence

- (a) $\frac{3}{5} = \frac{?}{100}$: Multiplier $= 100 \div 5 = 20 \implies 3 \times 20 = \mathbf{60}$. Fraction: $\frac{60}{100}$.
- (b) $\frac{4}{7} = \frac{28}{?}$: Multiplier $= 28 \div 4 = 7 \implies 7 \times 7 = \mathbf{49}$. Fraction: $\frac{28}{49}$.

2. Pages 142–144: Comparing, Adding & Subtracting Unlike Fractions

Comparing Fractions

- **Example 3 (Page 142): Compare $\frac{5}{6}$ and $\frac{7}{8}$:**
- Step 1: Find LCM of denominators 6 and 8. Prime factors: $6 = 2 \times 3$, $8 = 2^3$. $\text{LCM} = 24$.
- Step 2: Convert to like fractions:
 $\frac{5 \times 4}{6 \times 4} = \frac{20}{24}$, $\frac{7 \times 3}{8 \times 3} = \frac{21}{24}$.
- Step 3: Compare numerators: $20 < 21 \implies \mathbf{\frac{5}{6} < \frac{7}{8}}$.

Mental Maths / Speed Comparison (Page 143)

Train A travels $240\frac{1}{2}$ km in $3\frac{1}{2}$ hours. Train B travels $180\frac{1}{4}$ km in $2\frac{1}{2}$ hours. Which train is faster?

• Speed of Train A:

$$\text{Distance} = \frac{481}{2} \text{ km}, \text{ Time} = \frac{7}{2} \text{ hr}.$$

$$\text{Speed}_A = \frac{481}{2} \div \frac{7}{2} = \frac{481}{2} \times \frac{2}{7} = \frac{481}{7} = \mathbf{68.71 \text{ km/h}}.$$

• Speed of Train B:

$$\text{Distance} = \frac{721}{4} \text{ km}, \text{ Time} = \frac{5}{2} \text{ hr}.$$

$\text{Speed}_B = \frac{721}{4} \div \frac{5}{2} = \frac{721}{4} \times \frac{2}{5} = \frac{721}{10} = \mathbf{72.1 \text{ km/h}}$.

• **Conclusion:** Train B (72.1 km/h) is faster than Train A (68.71 km/h).

Example 7 (Page 143): Mixed Fraction Addition

Solve $3\frac{2}{5} + 2\frac{3}{4}$:

- Improper forms: $\frac{17}{5} + \frac{11}{4}$.
- LCM of 5 and 4 is 20:

$\frac{17 \times 4}{20} + \frac{11 \times 5}{20} = \frac{68}{20} + \frac{55}{20} = \frac{123}{20} = \mathbf{6\frac{3}{20}}$.

3. Pages 144–146: Simplification (HCF Method) & Multiplication of Fractions (Exercise 8.3)

Exercise 8.3 (Page 146)

Question 1: Multiply and Simplify to Lowest Form

- (a) $\frac{2}{3} \times \frac{5}{4}$:

$\frac{2 \times 5}{3 \times 4} = \frac{10}{12} = \mathbf{\frac{5}{6}}$.

- (b) $\frac{3}{8} \times \frac{4}{9}$:

Cancel cross common factors: $\frac{1}{2} \times \frac{1}{3} = \mathbf{\frac{1}{6}}$.

- (c) $2\frac{1}{3} \times 1\frac{1}{5}$:

Convert: $\frac{7}{3} \times \frac{6}{5} = \frac{7 \times 2}{1 \times 5} = \frac{14}{5} = \mathbf{2\frac{4}{5}}$.

Question 2: Convert into Mixed Fractions

- (a) $7 \times \frac{2}{3} = \frac{14}{3} = \mathbf{4\frac{2}{3}}$.
- (b) $4 \times \frac{3}{5} = \frac{12}{5} = \mathbf{2\frac{2}{5}}$.

Question 3: Area of Playground

Find the area of a rectangular playground that is $14\frac{2}{5} \text{ metres}$ long and $9\frac{5}{8} \text{ metres}$ wide.

- Length $= \frac{72}{5} \text{ m}$, Width $= \frac{77}{8} \text{ m}$.
- $\text{Area} = \frac{72}{5} \times \frac{77}{8} = \frac{9 \times 77}{5 \times 1} = \frac{693}{5} = \mathbf{138\frac{3}{5} \text{ m}^2}$ (or 138.6 m^2).

Question 4: Fractional Portions

- (a) $\frac{3}{8}$ of \$40: $\frac{3}{8} \times 40 = 3 \times 5 = \mathbf{15}$.
- (b) $\frac{2}{3}$ of 1 hour (in minutes): $\frac{2}{3} \times 60 \text{ min} = 2 \times 20 = \mathbf{40 \text{ minutes}}$.
- (c) $\frac{3}{4}$ of 2 kilograms (in grams): $\frac{3}{4} \times 2000 \text{ g} = 3 \times 500 = \mathbf{1500 \text{ grams}}$ (1.5 kg).

4. Pages 146–148: Division of Fractions, Reciprocals & Word Problems

Division Rules & Worked Examples

- **Division Rule (Stay, Change, Flip):** $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$.
- **Example 16 (Page 147):** A water tank contains 9 litres of water. It is poured into bottles of capacity $\frac{3}{4}$ litre. How many bottles can be filled?
 $\text{Number of bottles} = 9 \div \frac{3}{4} = 9 \times \frac{4}{3} = 3 \times 4 = 12$ bottles.
- **Example 18 (Page 147): Divide $5\frac{1}{4}$ by $2\frac{1}{3}$:**
 $\frac{21}{4} \div \frac{7}{3} = \frac{21}{4} \times \frac{3}{7} = \frac{3 \times 3}{4 \times 1} = \frac{9}{4} = 2\frac{1}{4}$.

Real-World Word Problems (Page 148)

Question 3

A rope of length $15\frac{3}{4}$ metres is cut into pieces of length $\frac{3}{4}$ metre each. Find the number of pieces.

- $\frac{63}{4} \div \frac{3}{4} = \frac{63}{4} \times \frac{4}{3} = 21$ pieces.

Question 4

A shopkeeper has 27 kg of dry fruits and distributes them equally among 36 children. How many grams does each child receive?

- Total in grams $= 27 \times 1000 = 27,000$ grams.
- Each child receives $= \frac{27000}{36} = 750$ grams (or $\frac{3}{4}$ kg).

Question 5

A garden is divided into 4 equal sections. $\frac{2}{3}$ of each section is planted with flowers. What fraction of the entire garden is planted with flowers? If the garden area is 240 m^2 , what is the planted area?

- Fraction of total garden $= 4 \times \left(\frac{1}{4} \times \frac{2}{3}\right) = \frac{2}{3}$.
- Planted area $= \frac{2}{3} \times 240 \text{ m}^2 = 160 \text{ m}^2$.

5. Pages 149–150: Chapter Drill — Section A: Multiple Choice Questions (MCQs 1–10)

MCQ 1

Which of the following is a unit fraction?

- (a) $\frac{4}{1}$ | (b) $\frac{1}{7}$ | (c) $\frac{7}{4}$ | (d) $\frac{4}{7}$

- **Correct Option: (b) $\frac{1}{7}$.**

MCQ 2

The reciprocal of $2\frac{1}{3}$ is:

(a) $\frac{7}{3}$ | (b) $\frac{3}{7}$ | (c) $\frac{1}{7}$ | (d) $2\frac{3}{1}$ *

• **Solution:** $2\frac{1}{3} = \frac{7}{3}$. The reciprocal is $\frac{3}{7}$.

• **Correct Option: (b) $\frac{3}{7}$.**

MCQ 3

To compare $\frac{2}{5}$ and $\frac{3}{4}$, the common denominator should be:

(a) 5 | (b) 4 | (c) 20 | (d) 10*

• **Correct Option: (c) 20.**

MCQ 4

A fraction is in its simplest form if its numerator and denominator are:

(a) Equal | (b) Co-prime | (c) Prime numbers | (d) Even numbers*

• **Correct Option: (b) Co-prime.**

MCQ 5

Compare $\frac{3}{7}$ and $\frac{5}{7}$:

(a) $>$ | (b) $<$ | (c) $=$ | (d) $\neq$ *

• **Correct Option: (b) $<$.**

MCQ 6

The value of $\frac{3}{4} \times \frac{4}{3}$ is:

(a) 0 | (b) 1 | (c) $\frac{12}{7}$ | (d) $\frac{7}{12}$ *

• **Correct Option: (b) 1.**

MCQ 7

An improper fraction is always:

(a) Less than 1 | (b) Greater than or equal to 1 | (c) Equal to 1 | (d) A unit fraction*

• **Correct Option: (b) Greater than or equal to 1.**

MCQ 8

$4 \div \frac{1}{2}$ is equal to:

(a) 2 | (b) 8 | (c) $\frac{1}{8}$ | (d) 4*

• **Solution:** $4 \times \frac{2}{1} = 8$.

• **Correct Option: (b) 8.**

MCQ 9

An equivalent fraction of $\frac{2}{3}$ with denominator 12 is:

(a) $\frac{4}{12}$ | (b) $\frac{8}{12}$ | (c) $\frac{6}{12}$ | (d) $\frac{10}{12}$ *

- **Correct Option: (b) $\frac{8}{12}$.**

MCQ 10

If $\frac{5}{8} = \frac{15}{x}$, then the value of x is:

(a) 16 | (b) 24 | (c) 30 | (d) 40

- **Solution:** $5 \times 3 = 15 \implies 8 \times 3 = 24$.

- **Correct Option: (b) 24.**

6. Page 150: Chapter Drill — Section B: Fill in the Blanks (Q1–Q6)

1. The number that does not have a reciprocal is **\$0\$**.
2. In the fraction $\frac{7}{11}$, the top number is called the **numerator**.
3. Fractions with the same denominator are called **like** fractions.
4. While dividing fractions, we multiply by the **reciprocal** of the divisor.
5. The product of $\frac{3}{5} \times \frac{5}{3}$ is **\$1\$**.
6. A fraction written as $3\frac{1}{2}$ is called a **mixed** fraction.

7. Pages 151–152: Chapter Drill — Section C & D: Very Short & Short Answer Questions

Very Short Answer Questions

1. **Define a proper fraction and give an example:**

A fraction whose numerator is strictly less than its denominator. *Example:* $\frac{3}{4}$.

2. **Convert $\frac{22}{5}$ into a mixed fraction:**

$22 \div 5 = 4$ quotient with remainder $2 \implies \mathbf{4\frac{2}{5}}$.

3. **Find the HCF of 24 and 36 and use it to simplify $\frac{24}{36}$:**

$\text{HCF}(24, 36) = 12$. Dividing both by 12: $\frac{24 \div 12}{36 \div 12} = \mathbf{\frac{2}{3}}$.

4. **State the reciprocal of $3\frac{1}{2}$:**

$3\frac{1}{2} = \frac{7}{2} \implies \text{Reciprocal} = \mathbf{\frac{2}{7}}$.

5. **Compare $\frac{5}{8}$ and $\frac{7}{12}$:**

Cross-multiply: $5 \times 12 = 60$; $8 \times 7 = 56$. Since $60 > 56$, $\mathbf{\frac{5}{8} > \frac{7}{12}}$.

6. **How many $\frac{1}{4}$ kg pieces can be cut from a cake of weight 4 kg?**

$4 \div \frac{1}{4} = 4 \times 4 = \mathbf{16}$ pieces.

7. **Which is heavier: $\frac{3}{5}$ of 500 grams or $\frac{1}{2}$ of 1 kg?**

- $\frac{3}{5} \times 500 = 300$ g.
- $\frac{1}{2} \times 1000 = 500$ g.

- $\frac{1}{2}$ of 1 kg is heavier ($500 \text{ g} > 300 \text{ g}$).

Short Answer Questions

1. Arrange $\frac{2}{3}$, $\frac{5}{6}$, $\frac{7}{12}$, $\frac{1}{2}$ in ascending order:

- Convert to common denominator 12:

$$\frac{2}{3} = \frac{8}{12}, \quad \frac{5}{6} = \frac{10}{12}, \quad \frac{7}{12} = \frac{7}{12}, \quad \frac{1}{2} = \frac{6}{12}.$$

- Ascending order: $\frac{6}{12} < \frac{7}{12} < \frac{8}{12} < \frac{10}{12} \implies \frac{1}{2} < \frac{7}{12} < \frac{2}{3} < \frac{5}{6}$.

2. Find x if $\frac{3}{7} = \frac{x}{35}$:

$$x = \frac{3 \times 35}{7} = 3 \times 5 = 15.$$

3. Convert $7\frac{2}{3}$ into an improper fraction:

$$7\frac{2}{3} = \frac{7 \times 3 + 2}{3} = \frac{23}{3}.$$

4. The weight of an object on the Moon is $\frac{1}{6}$ of its weight on Earth. If an object weighs 42 kg on Earth, what is its weight on the Moon?

$$\text{Weight on Moon} = \frac{1}{6} \times 42 = 7 \text{ kg}.$$

8. Pages 152–153: Chapter Drill — Section E: Long Answer Questions (Q1–Q10)

Question 1

A book has 120 pages and Rohan has read 80 pages. What fraction of the book has he read? Express in lowest form.

- Fraction $= \frac{80}{120} = \frac{8}{12} = \frac{2}{3}$.

Question 5

The area of a rectangular plot is $65\frac{1}{3} \text{ m}^2$. If its breadth is $5\frac{1}{3} \text{ metres}$, find its length.

- $\text{Length} = \frac{\text{Area}}{\text{Breadth}} = \frac{196\frac{1}{3}}{5\frac{1}{3}} = \frac{196\frac{1}{3} \times \frac{3}{16}}{5\frac{1}{3} \times \frac{3}{16}} = \frac{196\frac{1}{3} \times 3}{5\frac{1}{3} \times 3} = \frac{49\frac{1}{3}}{5\frac{1}{3}} = 12\frac{25}{32} \text{ m}.$

Question 6

A container holds 18 litres of juice. If small bottles of capacity $\frac{3}{4}$ litre each are filled, how many bottles are filled completely?

- $18 \div \frac{3}{4} = 18 \times \frac{4}{3} = 6 \times 4 = 24$ bottles.

Question 8: Vidya & Pratap Picnic Water Problem

Vidya and Pratap went for a picnic. Their water bottle contained 5 litres of water. Vidya consumed $\frac{2}{5}$ of the water. Pratap consumed the remaining water.

- (a) How much water did Vidya drink?

$$\text{Vidya} = \frac{2}{5} \times 5 \text{ L} = \mathbf{2 \text{ litres}}.$$

- (b) What fraction of the total water did Pratap drink?

$$\text{Pratap's fraction} = 1 - \frac{2}{5} = \mathbf{\frac{3}{5}}$$
 (which equals 3 litres).

Question 9: Sapling Distance Problem

Saili plants 4 saplings in a row. The distance between two adjacent saplings is $\frac{3}{4} \text{ m}$. Find the distance between the first and the last sapling.

- Number of gaps between 4 saplings in a row $= 4 - 1 = 3 \text{ gaps}$.
- Total distance $= 3 \times \frac{3}{4} \text{ m} = \frac{9}{4} \text{ m} = \mathbf{2\frac{1}{4} \text{ metres}}$ (2.25 m).

Question 10: Subject Preference Class Distribution

In a class of 40 students, $\frac{1}{5}$ like to study English, $\frac{2}{5}$ like to study Mathematics, and the remaining students like to study Science.

- (a) How many students like to study English?

$$\frac{1}{5} \times 40 = \mathbf{8 \text{ students}}.$$

- (b) How many students like to study Mathematics?

$$\frac{2}{5} \times 40 = \mathbf{16 \text{ students}}.$$

- (c) What fraction of students like to study Science?

$$\text{Science fraction} = 1 - \left(\frac{1}{5} + \frac{2}{5} \right) = 1 - \frac{3}{5} = \mathbf{\frac{2}{5}}$$
 (which equals 16 students).

9. Page 153: Competency-Based HOTS, Life Skills & Values

Higher Order Thinking Skills (HOTS)

Question 1

Find a fraction equivalent to $\frac{3}{5}$ where the product of the numerator and denominator is 135:

- Let the fraction be $\frac{3k}{5k}$.
- $(3k) \times (5k) = 135 \implies 15k^2 = 135 \implies k^2 = 9 \implies k = 3$.
- Numerator $= 3(3) = 9$, Denominator $= 5(3) = 15$.
- The fraction is $\frac{9}{15}$.

Question 2: Tap Filling Rate

If a tap fills $\frac{3}{7}$ of a tank in 1 hour, how much of the tank gets filled in $\frac{2}{3} \text{ hour}$?

$$\text{Fraction filled} = \frac{3}{7} \times \frac{2}{3} = \mathbf{\frac{2}{7} \text{ of the tank}}.$$

Life Skills & Values (Page 153)

Question 1

Why is it critical to use fractions with exact precision in medicine dosages, engineering construction, and culinary recipes?

- **Pedagogical Explanation:** In medicine, administering $\frac{1}{2}$ text{ mg} vs $\frac{1}{4}$ text{ mg} is a 100\% overdose error that could be lethal. In civil engineering, an error of $\frac{1}{8}$ text{ inch} in beam joint tolerances can compromise structural stability. Fractions ensure safety, reliability, and precision.

Question 2

How do fractions help us measure and minimize food waste and environmental pollution?

- **Pedagogical Explanation:** By calculating that $\frac{1}{3}$ of global food production is discarded, society can quantify carbon footprints and implement fractional inventory management to reduce resource depletion.

Final Pedagogical Summary of Chapter 8

Fractions are continuous proportions made tangible. Through 2D grid area multiplication, measuring-scoop division, and common denominator re-slicing, students understand that fractional arithmetic is fundamentally about physical conservation and scaling.
